OpenGL API's

(1) Initialization & setup :

- glutInit()
- glutCreateWindow("Title")
- glewInit()

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② Drawing primitives:

- glBegin (GL_TRIANGLES) --- glEND().
- glDrawArrays(), glDrawElements()

③ • Transformations:

- glTranslatef (x, y, z);
- glRotatef (Angle, x, y, z);
- glScalef (x, y, z).

④ Coloring and Texturing.

- glColor3f (r, g, b);
- glTexImage2D();

⑤ Shading and lighting.

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- glEnable (GL_lighting).

⑥ Buffer operations.

- glFlush(). displays the rendered image

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APIs for Rendering:

1. GL (OpenGL) : core graphics API.
2. GLU (OpenGL Utility Library) : Helper function for openGL
3. GLUT (OpenGL Utility Toolkit) : windowing & user input management.

(i) GL Functions

- glBegin(GL_TRIANGLES) ... glEnd() ; Defines primitive shapes
- glColor3f(r,g,b) ; sets colors
- glVertex3f(x,y,z) ; Defines a vertex
- glTranslatef(x,y,z)
- glRotatef(angle, x,y,z)
- glScalef(x,y,z)

(ii) GLU

- gluOrtho2D(left, right, bottom, top).
- gluPerspective(fov, aspect, near, far)
- gluLookAt(eyex, eyey, eyz, centerx, centery, centerz, upx, upy, upz)
- gluSphere( ), gluCylinder( )

(iii) GLUT:

- glutCreateWindow("OpenGL Window")
- glutDisplayFunc(displayFunc)
- glutKeyboardFunc(KeyboardFunc)
- glutMouseFunc(MouseFunc)
- glutMainLoop()

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lighting → glLightfv(light, pname, params)

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Call back functions

↳ Used via GLUT

- glutDisplayFunc(display)
- glutKeyboardFunc(Keyboard)
- glutReshapeFunc(reshape)

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